

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2022-23

Paper No. MTB-607: Number Theory

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) State the Peano axioms and the well-ordering principle for natural numbers. [3]
 - (b) Show that $\frac{a(a^2 + 2)}{3}$ is an integer for all $a \geq 1$. [2]
 - (c) Define the quadratic residue mod n . Show that Q_n is a subgroup of U_n . [3]
 - (d) Find the least non-negative residue of $3^{51} \pmod{23}$. [2]
 - (e) Solve the congruence $18k \equiv 42 \pmod{50}$ (if a solution exists). [2]
 - (f) Show that if (a, b, c) is a primitive Pythagorean triple, then exactly one of a and b is even and exactly one of them is divisible by 3. [2]
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Question 2.

- (a) If a and b are integers with $b \neq 0$, show that there exist unique integers q and r such that $a = qb + r$, where $0 \leq r < |b|$. [8]
 - (b) Establish the validity of the Euclidean algorithm. [6]
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Question 3.

- (a) Show that there exist arbitrarily large gaps between consecutive prime numbers. [4]
 - (b) Justify whether the following statements are True or False:
 - (i) There is a polynomial $f(x)$ with integer coefficients such that $f(n)$ is prime for every $n \in \mathbb{Z}$.
 - (ii) There is an arithmetic progression which consists solely of prime numbers.[8]
 - (c) State the Prime Number Theorem. [2]
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Question 4.

- (a) Let $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ are distinct primes. Show that $a \equiv b \pmod{n}$ if and only if $a \equiv b \pmod{p_i^{a_i}}$ for $i = 1, 2, \dots, k$. [4]
- (b) Use part (a) to express the congruence $13x \equiv 71 \pmod{380}$ as a system of congruences, and solve this system using the Chinese Remainder Theorem. [10]
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Question 5.

- (a) Show that an integer n is prime if and only if $(n-1)! \equiv -1 \pmod{n}$. [7]
- (b) Show that Euler's ϕ -function is multiplicative. [7]
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Question 6.

- (a) Show that for each positive integer $n \geq 1$: $n = \sum_{d|n} \phi(d)$. [7]
- (b) Show that there are no positive integer solutions x, y, z of $x^4 + y^4 = z^2$. Hence conclude Fermat's Last Theorem for $n = 4$. [7]
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Question 7. Let p be an odd prime. Show that:

- (a) If g is a primitive root mod p , then either g or $g + p$ is a primitive root mod p^2 . [7]
- (b) If h is a primitive root mod p^2 , then h is also a primitive root mod p^e for $e \geq 2$. [7]
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